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Substations and Lines (MPA-ESV 25/26L)

Task 3

Given Parameters

$S_n = 26 \text{ MVA}$, $i_{pn} = 50 \text{ kA}$, $I_{th} = 20 \text{ kA}$, $U_n = 6 \text{ kV}$, $t_k = 0.6 \text{ s}$
 $l = 0.8 \text{ m}$, $a_{min} = 0.09 \text{ m}$, $k_{th} \approx 135 \text{ A} \cdot \text{s}^{1/2}/\text{mm}^2$, $\sigma_{perm} \approx 160 \text{ MPa}$

1. Nominal Current Calculation

$$I_n = \frac{S_n}{\sqrt{3} \cdot U_n}$$
$$I_n = \frac{26 \times 10^6}{\sqrt{3} \cdot 6000} \approx 2501.8 \text{ A}$$

2. Selection of Busbars

Current per sub-conductor (dual busbar):

$$I_{n1} = \frac{I_n}{2} = \frac{2501.8}{2} = 1250.9 \text{ A}$$

Selected profile: Copper $100 \times 10 \text{ mm}$ ($A_1 = 1000 \text{ mm}^2$). Gap $a_s = 10 \text{ mm} = 0.01 \text{ m}$.
Total area:

$$A_{tot} = 2 \cdot A_1 = 2000 \text{ mm}^2$$

3. Thermal Short-Circuit Stability

$$A_{min} = \frac{I_{th} \cdot \sqrt{t_k}}{k_{th}}$$
$$A_{min} = \frac{20000 \cdot \sqrt{0.6}}{135} \approx 114.8 \text{ mm}^2$$

Check:

$$A_{tot} = 2000 \text{ mm}^2 \geq 114.8 \text{ mm}^2 \quad (\text{Stable})$$

4. Electrodynamic Short-Circuit Stability

4.1. Phase-to-Phase Forces

$$F_m = \frac{\sqrt{3}}{2} \cdot \frac{\mu_0}{2\pi} \cdot \frac{i_{pn}^2}{a_{min}} \cdot l$$
$$F_m = \frac{\sqrt{3}}{2} \cdot 2 \times 10^{-7} \cdot \frac{(50000)^2}{0.09} \cdot 0.8 \approx 3849 \text{ N}$$

Bending moment:

$$M = \frac{F_m \cdot l}{10} = \frac{3849 \cdot 0.8}{10} = 307.9 \text{ N} \cdot \text{m}$$

Section modulus (two bars, strong axis):

$$W = 2 \cdot \frac{h \cdot b^2}{6} = 2 \cdot \frac{10 \cdot 100^2}{6} = 33.33 \times 10^{-6} \text{ m}^3$$

Bending stress:

$$\sigma_m = \frac{M}{W} = \frac{307.9}{33.33 \times 10^{-6}} \approx 9.24 \text{ MPa}$$

Check: $\sigma_m < \sigma_{perm} \approx 160 \text{ MPa}$ (Stable)

4.2. Conductor-to-Conductor Forces (No spacers)

$$i_s = \frac{i_{pn}}{2} = \frac{50000}{2} = 25000 \text{ A}$$

Force ($l_s = l = 0.8 \text{ m}$):

$$F_s = \frac{\mu_0}{2\pi} \cdot \frac{i_s^2}{a_s} \cdot l_s = 2 \times 10^{-7} \cdot \frac{(25000)^2}{0.01} \cdot 0.8 = 10000 \text{ N}$$

Bending moment:

$$M_s = \frac{F_s \cdot l_s}{10} = \frac{10000 \cdot 0.8}{10} = 800 \text{ N} \cdot \text{m}$$

Section modulus (single bar, weak axis):

$$W_s = \frac{b \cdot h^2}{6} = \frac{100 \cdot 10^2}{6} = 1.667 \times 10^{-6} \text{ m}^3$$

Bending stress:

$$\sigma_s = \frac{M_s}{W_s} = \frac{800}{1.667 \times 10^{-6}} \approx 480 \text{ MPa}$$

Check: $\sigma_s > 160 \text{ MPa}$ (Fails)

4.3. Recalculation with One Spacer ($l_s = 0.4 \text{ m}$)

$$F_s = 2 \times 10^{-7} \cdot \frac{(25000)^2}{0.01} \cdot 0.4 = 5000 \text{ N}$$

$$M_s = \frac{5000 \cdot 0.4}{10} = 200 \text{ N} \cdot \text{m}$$

$$\sigma_s = \frac{200}{1.667 \times 10^{-6}} \approx 120 \text{ MPa}$$

Check: $\sigma_s \leq 160 \text{ MPa}$ (Stable)